



ARTIFICIAL INTELLIGENCE

Facebook is giving away its secrets so that it can better compete with Google—and everyone else.”

Deep learning can deeply impact our lives

Deep learning's applications range from medical diagnosis to marketing, and we are not kidding you. There is a fabulous line on IBM's website, which says that we are all experiencing the benefits of deep learning today, in some way or the other, without even realising it.

In June 2016, Ford researchers announced that they had developed a very accurate approach to estimate a moving vehicle's position within a lane in real time. They achieved this kind of sub-centimetre-level precision by training a DNN, which they call DeepLanes, to process input images from two laterally-mounted down-facing cameras—each recording at an average 100 frames/s.

They trained the neural network on an NVIDIA DIGITS DevBox with cuDNN-accelerated Caffe deep learning framework. NVIDIA DIGITS is an interactive workflow based solution for image classification. NVIDIA's software development kit has several powerful tools and libraries for developing deep learning frameworks, including Caffe, CNTK, TensorFlow, Theano and Torch.

The life sciences industry uses deep learning extensively for drug discovery, understanding of disease progression and so on. Researchers at The Australian National University, for example, are using deep learning to understand the progression of Parkinson's disease. In September, researchers at Duke University revealed a method that uses deep learning and light based, holographic scans to spot malaria-infected cells in a simple blood sample, without human intervention.

Abu Qader, a high school student in Chicago, has created GliLab, a startup that combines AI with the findings of mammograms and fine-needle aspirations to identify and classify breast cancer tumours. The solution starts with mammogram imaging and then sifts Big Data to build predictive models about similar tumour types, risks, growth, treatment outcomes and so on. He used an NVIDIA GeForce GT 750M GPU on his laptop along with TensorFlow deep learning framework.

Deep learning is supposed to be the future of digital personal assistants like Siri, Alexa and Cortana. Bark out any command, and these personal assistants will be able to understand and get it done. Deep learning is also going to be the future of Web search, marketing, product design, life sciences and much more.

Once the Internet of Things (IoT) ensnares the world in its Web, there is going to be Big(ger) Data for deep learning systems to work on. No wonder companies ranging from Google, Facebook, Microsoft and Amazon, to NVIDIA, Apple, AMD and IBM, are all hell-bent on leading the deep learning race. A year down the line, we will have a lot more to talk about! **EFY**

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